

Lecture Notes on Differential Equations and Transform Techniques

Lecture Notes for Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 01****Topics covered :**

1. Introduction to Differential Equations

- Dynamic modeling of various phenomena across multiple domains, such as physics, chemistry and economics (most importantly, electrical engineering in the context of this course) often involves differential equations. On the other hand, steady steady state modeling of various phenomena often involves system of linear/non-linear equations.
- An equation involving one dependent variable (or a function) [?] and its derivatives with respect to one or more independent variables is called a differential equation.
- For a function of a single variable $y = f(x) : \mathbb{R} \rightarrow \mathbb{R}$, the derivative of $f(x)$ at x^0 is defined as (1) and represents the rate of change of the function at x^0 .

$$\left. \frac{dy}{dx} (\text{or } y' \text{ or } f'(x)) \right|_{x^0} = \lim_{x \rightarrow x^0} \frac{f(x) - f(x^0)}{x - x^0} \quad (1)$$

The derivative of the function $f(x)$ at x^0 also represents the slope of the tangent line to the curve $f(x)$ at x^0 . This concept is fundamental for numerically solving differential equations.

A differential equation involving derivatives of a function of a single variable is called the **ordinary differential equation (ODE)**.

- For a function of multiple variables $y = f(\mathbf{x}) : \mathbb{R}^N \rightarrow \mathbb{R}$, the partial derivative of $f(\mathbf{x})$ with respect to a variable (say x_i) at $\mathbf{x}^0$ is defined as

$$\left. \frac{\partial y}{\partial x_i} \right|_{\mathbf{x}^0} = \lim_{h \rightarrow 0} \frac{f(x_1^0, \dots, x_i^0 + h, \dots, x_n^0) - f(\mathbf{x}^0)}{h} = \lim_{h \rightarrow 0} \frac{f(\mathbf{x}^0 + h\mathbf{e}_i) - f(\mathbf{x}^0)}{h} \quad (2)$$

The partial derivative of $f(\mathbf{x})$ with respect to a variable (say x_i) represents the rate of change of the function with respect to x_i while holding the other variables constant. In other words, the partial derivative measures how $f(\mathbf{x})$ changes as x_i is varied, while all other components of $\mathbf{x}$ are kept fixed.

The vector of partial derivatives of a function (with respect to all the variables) is called the **gradient** ($\nabla f(\mathbf{x})$).

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_i} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{bmatrix} \quad (3)$$

A differential equation that involves partial derivatives of a function of multiple variables is called a **partial differential equation (PDE)**.

- When solving differential equations, the objective is to find a function $f(x)$ (in case of ODEs) or $f(\mathbf{x})$ (in case of PDEs) that satisfies the given equation.